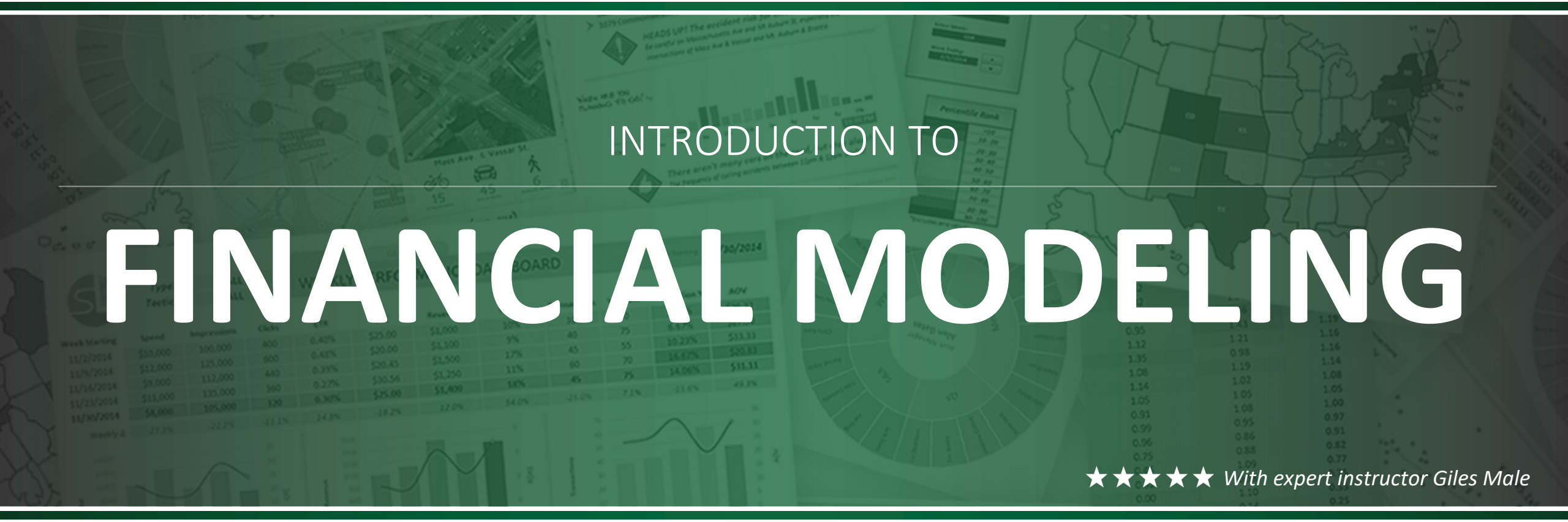


INTRODUCTION TO

FINANCIAL MODELING

★★★★★ *With expert instructor Giles Male*

Full Stack Modeller

MAVEN ANALYTICS

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COURSE OUTLINE

1 **Financial Modelling 101**	*Understand who builds financial models, why they are built, what types of models exist, and how they are typically developed*
2 **Standards & Best Practices**	*Explore the most prominent financial modelling standards, and some generally-accepted best practices and modelling principles*
3 **CASE STUDY:** Basic Revenue Forecast	*Create a simple revenue forecast model, while applying common modelling standards and best practices*
4 **Flows & Balances**	*Understand the difference between modelling flows and balances, and how these elements are connected to create financial models*
5 **CASE STUDY:** CAPEX Flows & Balances	*Model all relevant flows and balances for Sarah's Bakery as she looks to invest money in fixed assets*
6 **Three Statement Modelling**	*Learn the key technical principles behind three statement modelling, which forms the foundation for more advanced financial models*
7 **CASE STUDY:** Three Statement Model	*Link the three statements for Sarah's Bakery, and review the model to identify key patterns and insights*
8 **Recap & Next Steps**	*Highlight the key takeaways from this course, and introduce additional learning opportunities for those who want to dive deeper*

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THE COURSE PROJECT

THE SITUATION

You've just been hired as a Financial Modelling Consultant for **Sarah's Bakery**, a small, independently owned bakery operating on the streets of London

THE BRIEF

Sarah's baked goods are loved by the locals, and she is eager to expand her business by leasing a larger, brick-and-mortar space.

Your role is to help Sarah build some basic models to assess the financial health and potential profitability of her business, including a simple revenue projection, a capital expenditure forecast, and a complete three-statement model.

THE OBJECTIVES

- Review and understand core assumptions provided by Sarah
- Create consistent and easy-to-follow forecast models
- Follow generally-accepted principles and best practices
- Help Sarah uncover key insights about the bakery's finances



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SETTING EXPECTATIONS



This course is designed for **financial modelling beginners**

- *If you are looking for advanced techniques or industry-specific training, this course isn't the right fit for you; we will introduce common models and best practices, and keep things at an introductory level*



Our goal is to introduce **fundamental financial modelling principles**

- *Our goal is to help you build confidence and basic modelling literacy; we'll cover core financial modelling skills, but will not practice building complex models from scratch*



We'll cover a mix of **theory** and **practical skills**

- *This course will help you build a broad understanding of the financial modelling space at a high level before diving into common model types and practical exercises*



We'll use **Excel** for **Office** 365 on a **PC** for course assignments and demos

- *You should have a good working knowledge of Excel features and functions (note that what you see on your screen may not always match what you see on mine, especially if you are running a different version of Excel)*

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FINANCIAL MODELLING 101

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FINANCIAL MODELLING 101



In this section we'll review the broader financial modelling landscape, introduce general frameworks and definitions, identify key players, and explore common types of models

TOPICS WE'LL COVER:

- Intro to Modelling
- General Definitions
- Key Players
- Business Lifecycle
- Model Types
- Key Takeaways

GOALS FOR THIS SECTION:

- *Explain why financial modelling is worth learning*
- *Introduce key questions for building effective models*
- *Review common definitions of financial modelling*
- *Highlight key players, and the role they each play in the business lifecycle*
- *Introduce some of the most common types of financial models, and highlight key similarities and differences*

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THE SINGLE MOST IMPORTANT MESSAGE

Intro to Modelling

General Definitions

Key Players

Business Lifecycle

Model Types

Key Takeaways

FINANCIAL MODELS COME IN ALL SHAPES AND SIZES

- Ask 10 professional financial modellers the same question, and you may get **10 different responses**
- Remember that not all good models look the same, and that **bad models can be well-disguised**
- There's no single "right" way to build a model, but there are **general principles and practices** that you can apply to create highly effective financial models



Throughout this course we'll share proven frameworks and best practices that you can apply to any type of financial model.

Our goal is to help you build financial modelling literacy and begin to develop core foundational skills.

"All models are wrong. Some are useful" *- George Box*

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WHY LEARN FINANCIAL MODELLING?

- Intro to Modelling
- General Definitions
- Key Players
- Business Lifecycle
- Model Types
- Key Takeaways



Financial modelling provides **important and unique perspectives into business performance**, including financial health, growth potential, valuation, cash flow, and more



Even a basic level of financial literacy can help you understand and interpret common models and financial statements, and **discover the stories and insights hidden behind the data**

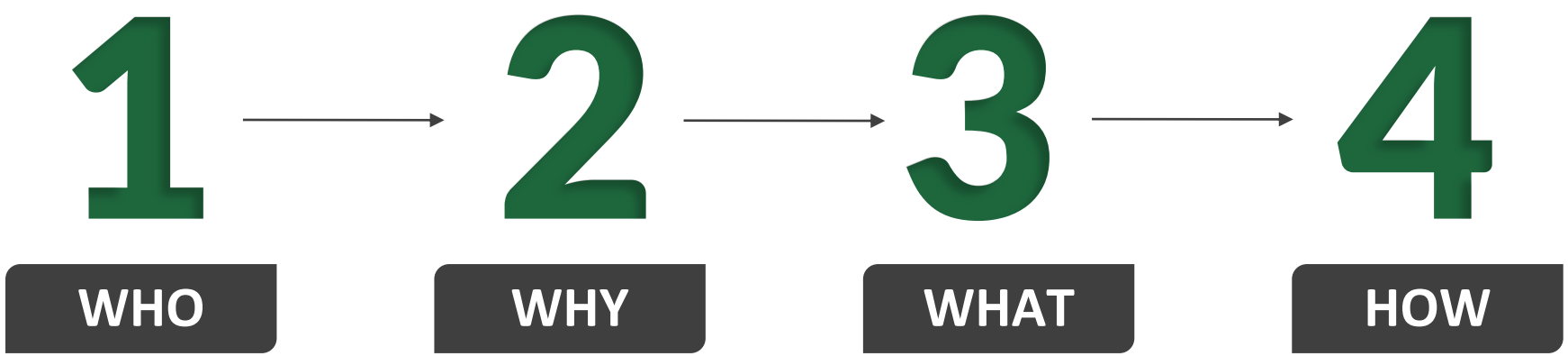


Financial modelling skills are highly versatile, relevant to finance and non-finance professionals alike, and **add value to virtually any team**

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THE FOUR KEY QUESTIONS

- Intro to Modelling
- General Definitions
- Key Players
- Business Lifecycle
- Model Types
- Key Takeaways



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THE FOUR KEY QUESTIONS

Intro to Modelling

General Definitions

Key Players

Business Lifecycle

Model Types

Key Takeaways

1 → **2** → **3** → **4**

WHO | **WHY** | **WHAT** | **HOW**

- ***Who*** *is the project sponsor or key stakeholder?*
- ***Who*** *is responsible for developing and maintaining the model?*
- ***Who*** *are the end users viewing or interacting with the model?*

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THE FOUR KEY QUESTIONS

Intro to Modelling

General Definitions

Key Players

Business Lifecycle

Model Types

Key Takeaways

1 → **2** → 3 → 4

WHO | **WHY** | WHAT | HOW

- ***Why*** *is the model needed?*
- *What core purpose will the model serve for the business?*
- *Which specific questions with the model answer, or NOT answer?*

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THE FOUR KEY QUESTIONS

Intro to Modelling

General Definitions

Key Players

Business Lifecycle

Model Types

Key Takeaways

1 → 2 → **3** → 4

WHO | WHY | **WHAT** | HOW

- ***What*** *type of model is most appropriate?*
- ***What*** *specific outputs or KPIs should the model produce?*
- ***What*** *type of parameters or assumptions are required?*

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THE FOUR KEY QUESTIONS

- Intro to Modelling
- General Definitions
- Key Players
- Business Lifecycle
- Model Types
- Key Takeaways

1 → 2 → 3 → **4**

WHO | WHY | WHAT | **HOW**

- ***How*** *should the model be built?*
- ***How*** *will you structure your model layout and components?*
- ***How*** *will the model be used by and shared with end users?*

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GENERAL DEFINITIONS

Intro to Modelling

General Definitions

Key Players

Business Lifecycle

Model Types

Key Takeaways

FINANCIAL MODEL *noun*

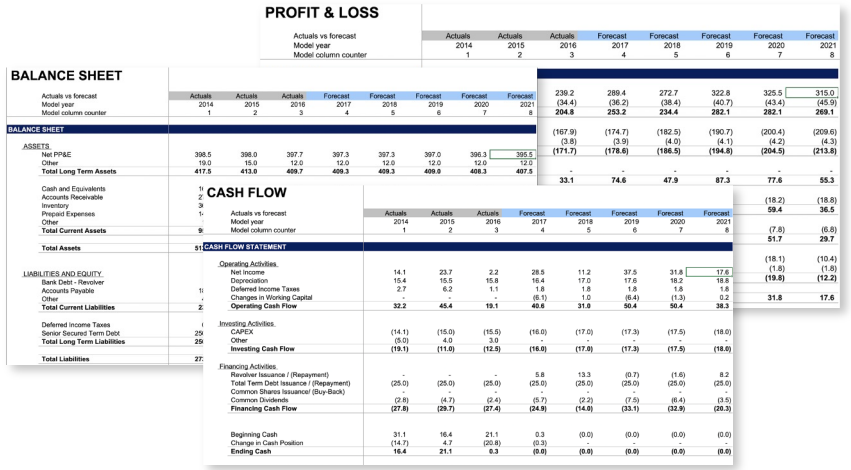
A time-based representation of an entity or project's uncertain, future financial performance

Regardless of the definition, all financial models should:

- Help stakeholders make informed, data-driven business decisions
- Serve a clear purpose, and answer a specific question or set of questions

Common elements:

- Inputs, calculations, outputs
- Controls
- Timelines
- Flows & balances
- Income Statement (Profit & Loss), Balance Sheet, Cash Flow Statement



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KEY PLAYERS

Intro to Modelling

General Definitions

Key Players

Business Lifecycle

Model Types

Key Takeaways

There are several types of **key players** or entities in the financial modelling space, from VC and private equity firms to accounting and internal finance teams

Venture Capital

A form of private investment focused on provisioning funding for early-stage companies with high growth potential

Private Equity

A latter-stage private investment with a focus on strategic management (often through buyout, leveraging debt)

Investment Banks

Advisers, traders, research houses, and intermediaries for complex financial transactions like IPOs and M&A

Project Finance

Funding of longer-term projects where debt funding is significant, and repayments are "sculpted" in line with expected cash flows

Accounting

Public accounting firms supporting businesses with financial statement preparations, tax returns, advisory services and more

Internal Finance Teams

Finance teams responsible for the ongoing financial management of the company (FP&A, Treasury, Accounting, Corporate Finance, etc.)

This is where a LOT of financial modelling happens!

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THE BUSINESS LIFECYCLE

Intro to Modelling

General Definitions

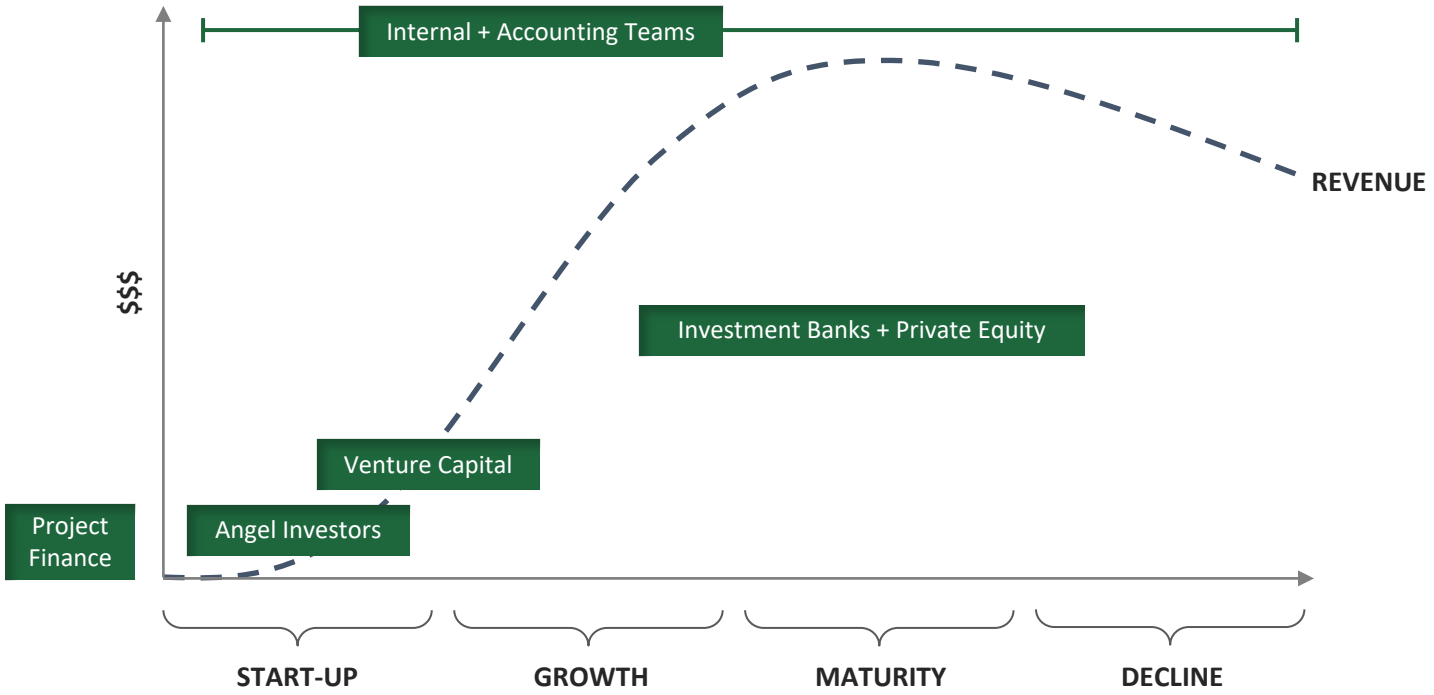
Key Players

Business Lifecycle

Model Types

Key Takeaways

Key players get involved at various stages of the **business lifecycle**, from start-up and growth through maturity and decline



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THE BUSINESS LIFECYCLE

Intro to Modelling

General Definitions

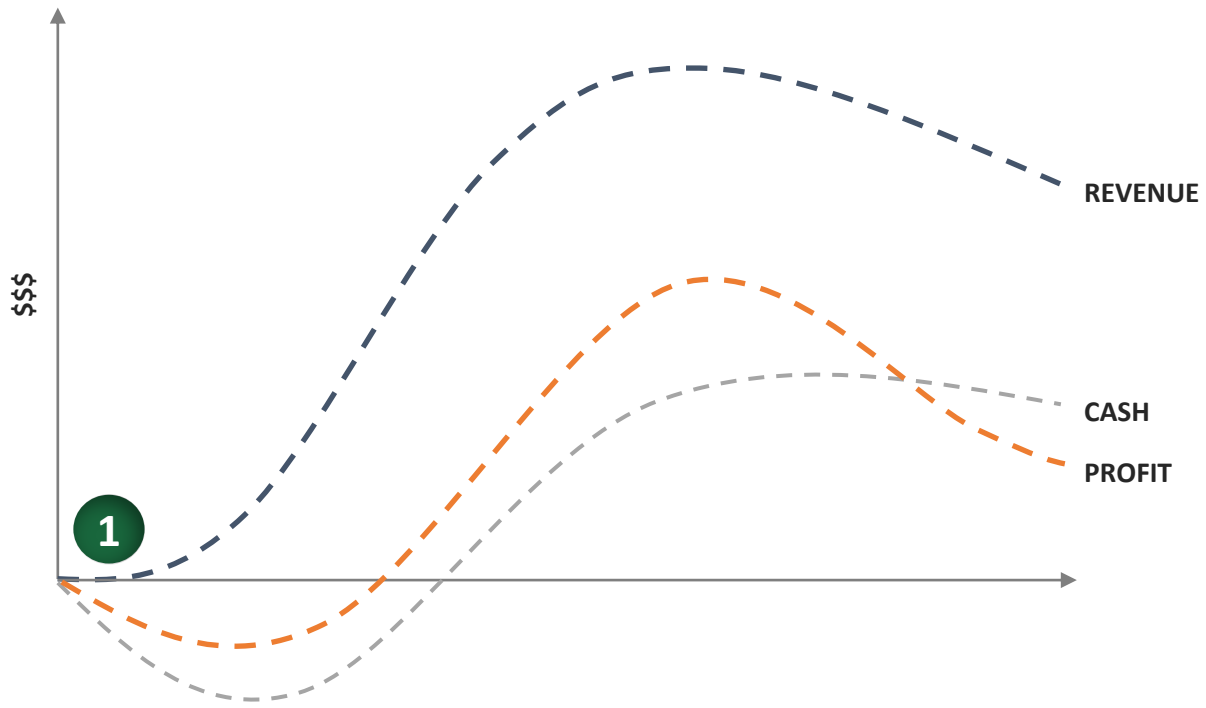
Key Players

Business Lifecycle

Model Types

Key Takeaways

The models, intentions, outputs and KPIs can vary significantly depending on the company, lifecycle stage, and team or entity involved



1

In the very early start-up phase, you may see high-level financial modelling as part of the initial business planning process.

At this point, it's common to see models focusing on defining the addressable market or forecasting short-term cash flow.

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THE BUSINESS LIFECYCLE

Intro to Modelling

General Definitions

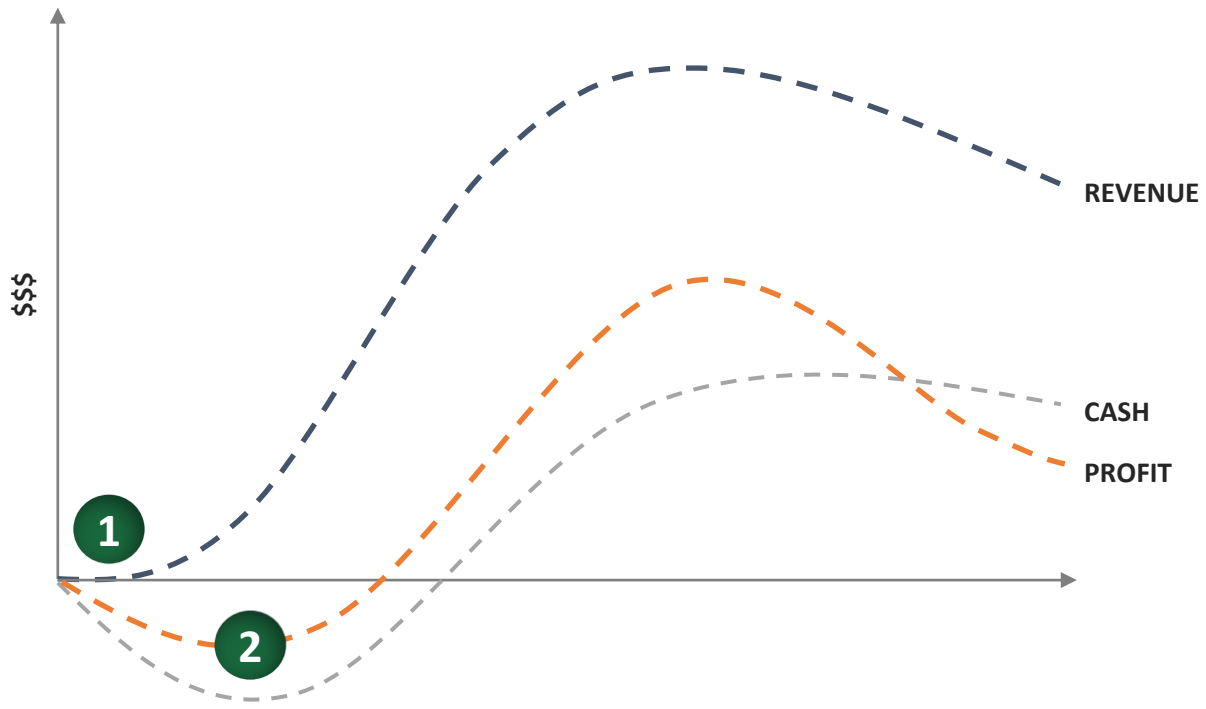
Key Players

Business Lifecycle

Model Types

Key Takeaways

The models, intentions, outputs and KPIs can vary significantly depending on the company, lifecycle stage, and team or entity involved



2

At this stage, the business is operational and (likely) has paying customers. At this point it's common to see basic accounting processes like budgeting and forecasting.

If investors are involved, you can expect to see more comprehensive forecast models like scenario plans and ROI analyses.

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THE BUSINESS LIFECYCLE

Intro to Modelling

General Definitions

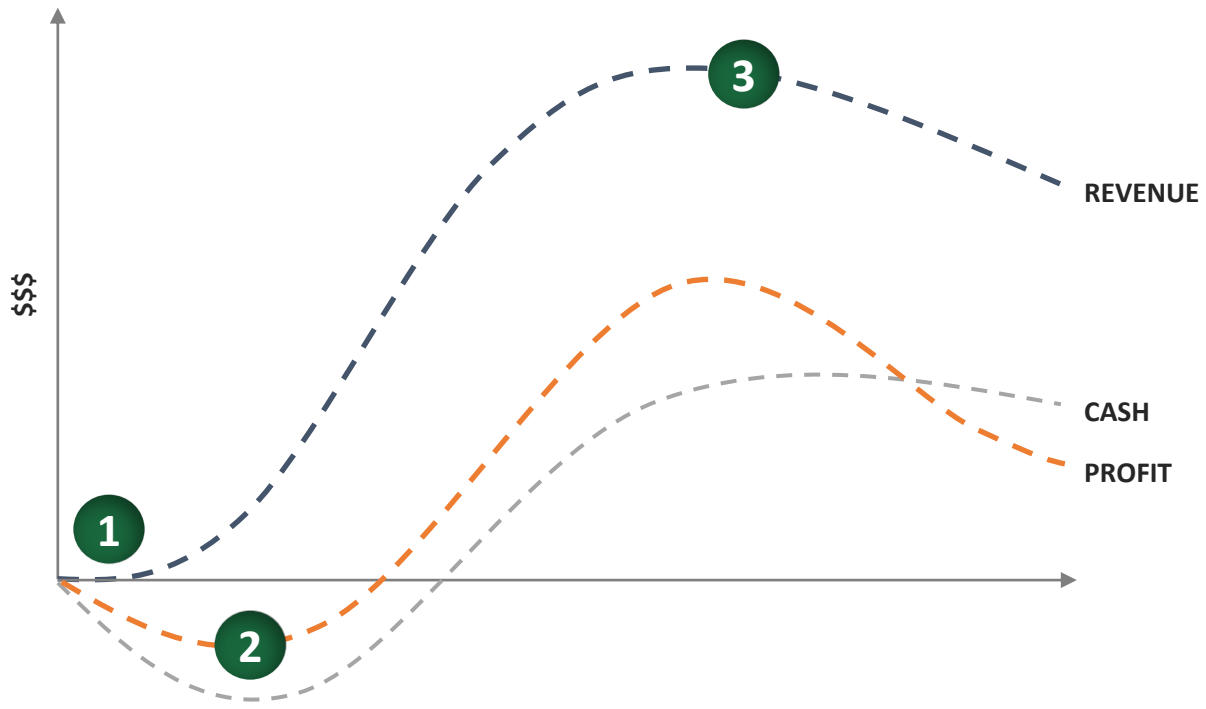
Key Players

Business Lifecycle

Model Types

Key Takeaways

The models, intentions, outputs and KPIs can vary significantly depending on the company, lifecycle stage, and team or entity involved



3

At maturity, most companies have a dedicated finance function owning a variety of models to manage routine operations plus central and departmental forecasting.

More complex models are common for companies engaging in larger strategic initiatives like an IPO, merger, acquisition, debt raise, etc.

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COMMON MODEL TYPES

Intro to Modelling

General Definitions

Key Players

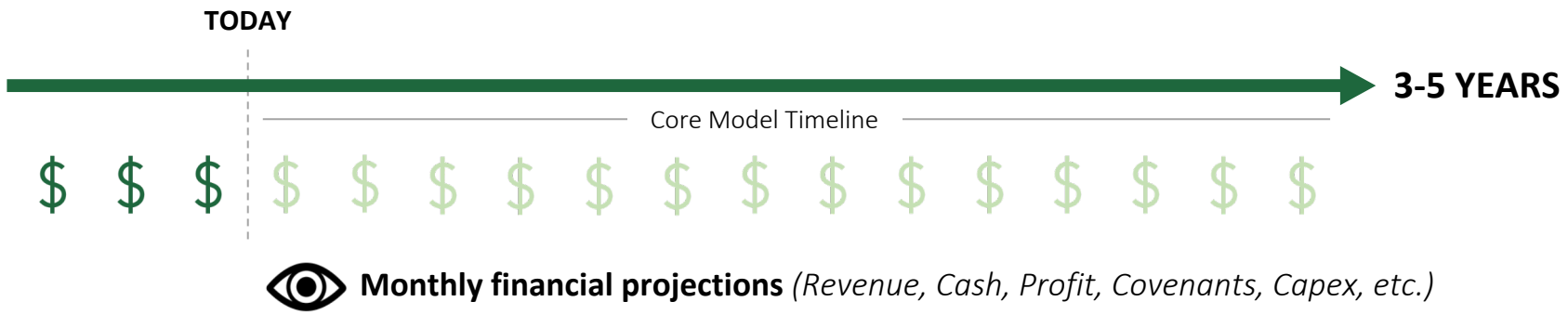
Business Lifecycle

Model Types

Key Takeaways

INTERNAL FORECAST

- Commonly used to **assess the financial impact of specific events or strategic decisions**
- Provides financial projections to help business managers and team leaders plan, typically at a **monthly-level** over a **3 to 5-year** forecast period
- Often developed by or linked closely to commercial or FP&A teams



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COMMON MODEL TYPES

Intro to Modelling

General Definitions

Key Players

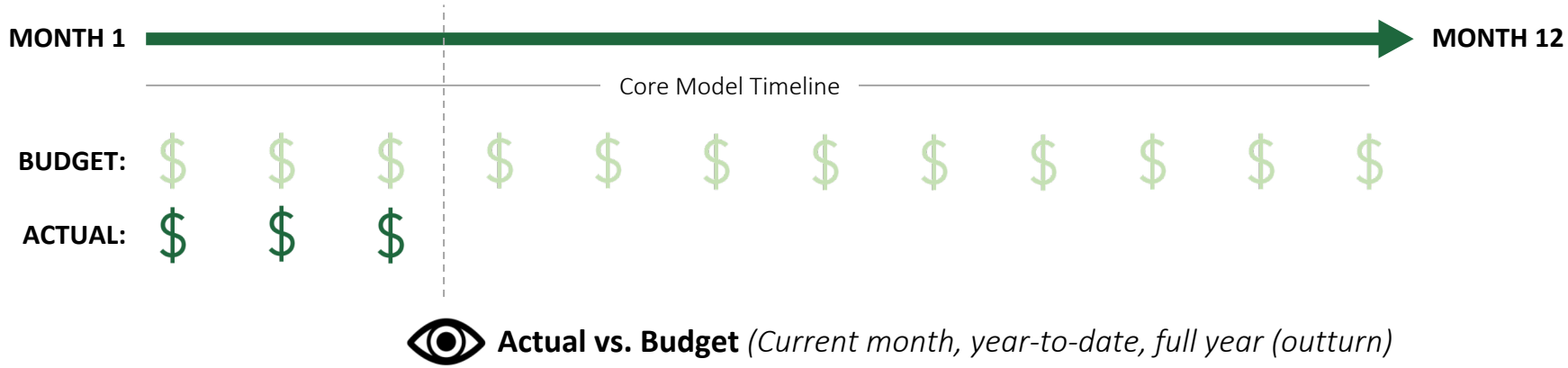
Business Lifecycle

Model Types

Key Takeaways

ACTUALS VS. BUDGET

- Commonly used to **compare actual business metrics against expectations**
- Provides monthly budget forecasts and actuals, typically over a **12-month timeline**
- Often based on Income Statement (P&L) performance, with heavier data use



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COMMON MODEL TYPES

Intro to Modelling

General Definitions

Key Players

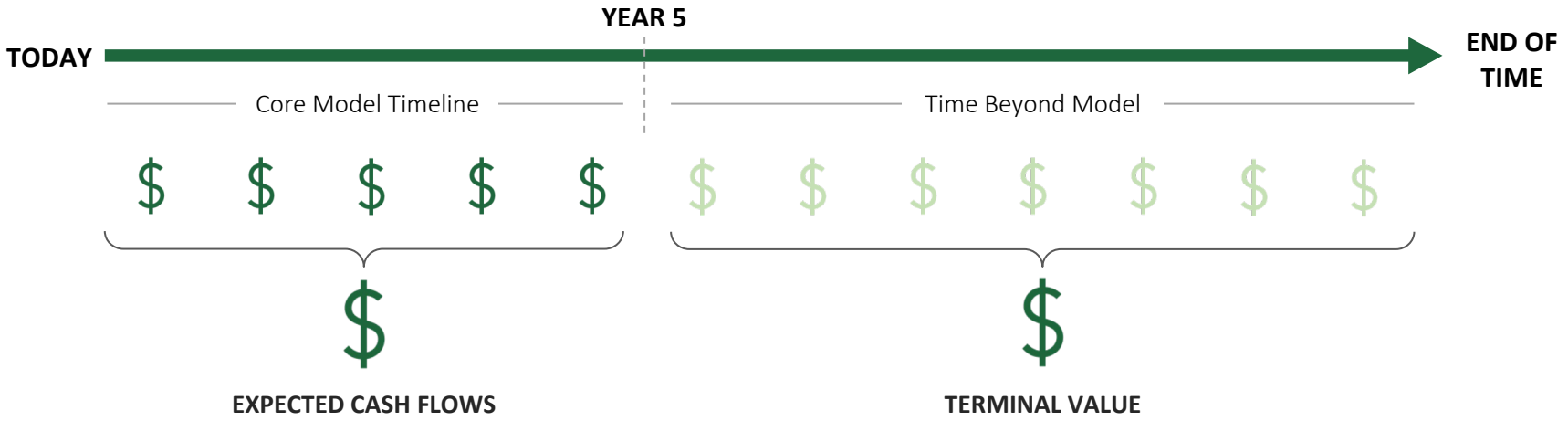
Business Lifecycle

Model Types

Key Takeaways

VALUATION MODEL

- Commonly used to **estimate Enterprise or Equity value**
- Often based on past performance plus growth assumptions
- **Discounted Cash Flow (DCF)** valuation models are used to assess investments based on expected future cash flows (typically over a **5 to 8-year period**) and a **Terminal Value**, which is the value of all cash flows beyond the core model timeline



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COMMON MODEL TYPES

Intro to Modelling

General Definitions

Key Players

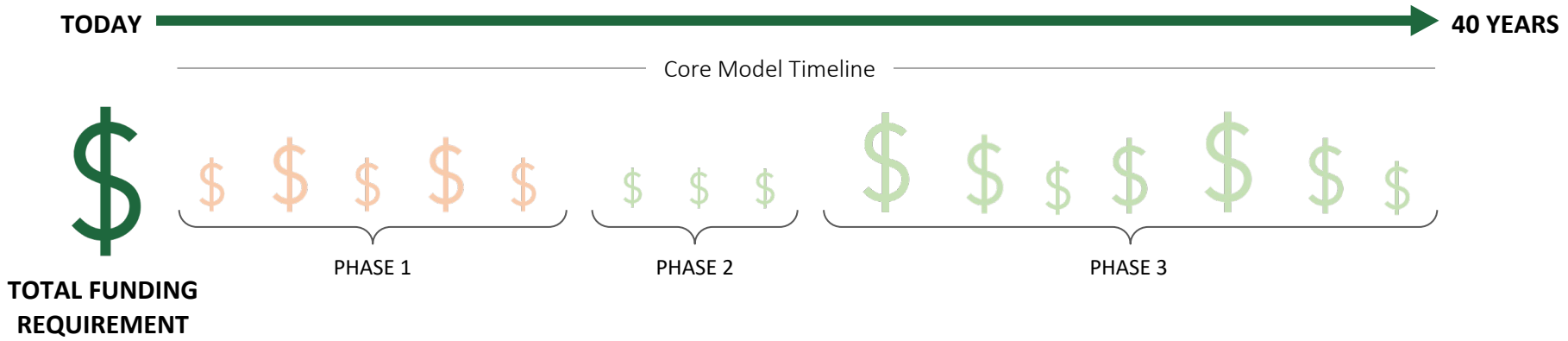
Business Lifecycle

Model Types

Key Takeaways

PROJECT FINANCE MODEL

- Commonly used to **assess the financial feasibility of a project or structure significant financing** (known as “debt sculpting”)
- Often focused on long-term cash flows, with distinct model and project phases
- Typically forward-looking, with no historic data



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KEY TAKEAWAYS

Intro to Modelling

General Definitions

Key Players

Business Lifecycle

Model Types

Key Takeaways

- Financial modelling skills are **highly versatile**, and provide an important perspective into a company or project's present and future financial health
- Remember the **4 key questions: who** are the stakeholders, **why** is the model needed, **what** type is required, and **how** should you develop it?
- There are several key players in the financial modelling world, including **accounting firms, VCs, investment banks** and **internal finance teams**
- Models **vary in scope, purpose and complexity**, depending on the business, lifecycle stage, and team or entity involved
- While definitions vary, all financial models should **serve a clear purpose** and **help stakeholders make informed, data-driven decisions**

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STANDARDS & BEST PRACTICES

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STANDARDS & BEST PRACTICES



In this section we'll review core financial modelling principles and industry standards, introduce the topic of periodicity, and address some common modelling mistakes

TOPICS WE'LL COVER:

- **Industry Standards**
- **ICAEW 20 Principles**
- **Universal Principles**
- **Periodicity**
- **Common Mistakes**
- **Key Takeaways**

GOALS FOR THIS SECTION:

- *Introduce common industry standards, including the ICAEW's "20 Principles for Good Spreadsheet Practice"*
- *Review some of the most universally-accepted principles*
- *Highlight the importance of periodicity, and how it can be integrated into various types of financial models*
- *Address some of the most common modelling errors*

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INDUSTRY STANDARDS

Industry Standards

ICAEW 20 Principles

Universal Principles

Periodicity

Common Mistakes

Key Takeaways

Industry standards are **sets of principles or rules** designed to raise the overall standard of financial modelling

- There are multiple standards and governing bodies, including the **Institute of Chartered Accountants in England and Wales (ICAEW),** the **FAST Standard** (Flexible, Appropriate, Structured, Transparent), **Best Practice Modelling (BPM)** and more



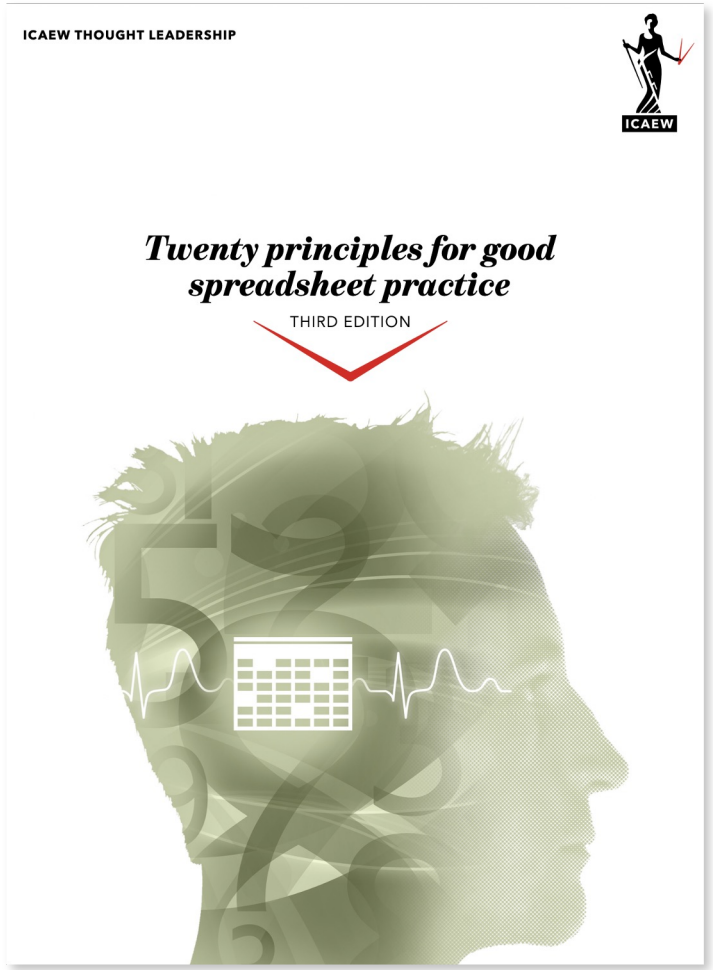


Even among professional modellers, ***there are various (often opposing) perspectives & views***

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ICAEW 20 PRINCIPLES

- Industry Standards
- **ICAEW 20 Principles**
- Universal Principles
- Periodicity
- Common Mistakes
- Key Takeaways



ICAEW's "Twenty Principles for Good Spreadsheet Practice" is an excellent starting point to learn common modelling principles

- ✓ Principle-based
- ✓ Clear and simple
- ✓ Covers all key fundamentals
- ✓ Free to download from **icaew.com**

ASSIGNMENT:

Download and read the full **ICAEW 20 Principles** guide before moving ahead to the next lesson

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UNIVERSALLY AGREED PRINCIPLES

- Industry Standards
- ICAEW 20 Principles
- **Universal Principles**
- Periodicity
- Common Mistakes
- Key Takeaways

- Be **consistent** with your model structure and calculation logic
- Keep your models as **clear and simple** as possible
- **Separate and distinctly format** your inputs, calculations and outputs
- Ensure that you have a **clear and approved scope** for your model
- Ensure that your model is **comprehensively tested** prior to use

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PERIODICITY

Industry Standards

ICAEW 20 Principles

Universal Principles

Periodicity

Common Mistakes

Key Takeaways

Periodicity describes the time periods or intervals captured in a financial model (monthly, annual, quarterly, semi-annual, etc.)

- Periodicity is typically managed through a dedicated “Time” tab, but can take many forms and require varying levels of complexity depending on the model context

Actuals / forecast	Actuals	Forecast	Forecast	Forecast	Forecast	Forecast	Forecast	Forecast
Model year	2024	2025	2026	2027	2028	2029	2030	2031
Model column counter	1	2	3	4	5	6	7	8

In its most basic form, the periodicity of a model could be a short series of hardcoded values / dates

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PERIODICITY

Industry Standards

ICAEW 20 Principles

Universal Principles

Periodicity

Common Mistakes

Key Takeaways

Periodicity describes the time periods or intervals captured in a financial model (monthly, annual, quarterly, semi-annual, etc.)

- Periodicity is typically managed through a dedicated "Time" tab, but can take many forms and require varying levels of complexity depending on the model context

	E	F	G	H	J	K	L	M	N	O	P
1	**Time**										
2	Model period ending				**31 May 21**	**30 Jun 21**	**31 Jul 21**	**31 Aug 21**	**30 Sep 21**	**31 Oct 21**	**30 Nov 21**
3	Pre-forecast vs forecast				Pre-forecast	Forecast	Forecast	Forecast	Forecast	Forecast	Forecast
4	Financial year ending				2021	2021	2022	2022	2022	2022	2022
5	Model column counter	**Constant**	**Unit**	**Total**	1	2	3	4	5	6	7
6											
7	**TIME RULER**										
8											
9	Model column counter		counter		1.0	2.0	3.0	4.0	5.0	6.0	7.0
10	First model column flag		flag	1	1	-	-	-	-	-	-
11											
12	1st model column start date	01 May 2021	date								
13	Months per model period	1	months								
14	First model column flag	-	flag	1 -	1	-	-	-	-	-	-
15	Model period beginning		date		01 May 21	01 Jun 21	01 Jul 21	01 Aug 21	01 Sep 21	01 Oct 21	01 Nov 21
16											
17	Months per model period	1	months								
18	Model period beginning	-	date	- -	01 May 21	01 Jun 21	01 Jul 21	01 Aug 21	01 Sep 21	01 Oct 21	01 Nov 21
19	Model period ending		date		31 May 21	30 Jun 21	31 Jul 21	31 Aug 21	30 Sep 21	31 Oct 21	30 Nov 21
20											
21	First modelling column financial year number	2021	year #								
22	Financial year end month number	6	month #								
23	Model period ending	-	date	- -	31 May 21	30 Jun 21	31 Jul 21	31 Aug 21	30 Sep 21	31 Oct 21	30 Nov 21
24	First model column flag	-	flag	1 -	1	-	-	-	-	-	-
25	Financial year ending		year #		2021	2021	2022	2022	2022	2022	2022
26											
27											
28	**FLAGS**										
29											
30	**Acquisition / initial balance date flag**										
31	Opening balance sheet date	31 May 2021	date								
32	Model period ending	-	date	- -	31 May 21	30 Jun 21	31 Jul 21	31 Aug 21	30 Sep 21	31 Oct 21	30 Nov 21
33	Acquisition / initial balance date flag		flag	1	1	-	-	-	-	-	-
34											
35	**Forecast period flag**										
36	Opening balance sheet date	31 May 2021	date								
37	Length of forecast period	5	years								
38	Last forecast date	31 May 2026	date								
39											
40	Opening balance sheet date	31 May 2021	date								
41	Last forecast date	31 May 2026	date								
42	Model period ending	-	date	- -	31 May 21	30 Jun 21	31 Jul 21	31 Aug 21	30 Sep 21	31 Oct 21	30 Nov 21
43	Forecast period flag		flag	60	-	1	1	1	1	1	1
44											
45	**Last forecast period flag**										
46	Last forecast date	31 May 2026	date								
47	Model period ending	-	date	- -	31 May 21	30 Jun 21	31 Jul 21	31 Aug 21	30 Sep 21	31 Oct 21	30 Nov 21
48	Last forecast period flag		flag	1	-	-	-	-	-	-	-
49											

Alternatively, it could be a dedicated, complex worksheet containing all time-related logic (fiscal calendars, time periods, pre/post forecast dates, etc.)

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COMMON MISTAKES

- Industry Standards
- ICAEW 20 Principles
- Universal Principles
- Periodicity
- **Common Mistakes**
- Key Takeaways



Failing to see the **purpose or bigger picture** (scope, commerciality, materiality)



Not following **rules or universal standards**



Trying to solve **every modelling challenge** at once



Believing that your way is the **best or only way**



Expecting everything to fit **known or existing templates**



Failing to stay up to date on **emerging tools and technologies**



Failing to **thoroughly test and QA** models before sharing with end users

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KEY TAKEAWAYS

- Industry Standards
- ICAEW 20 Principles
- Universal Principles
- Periodicity
- Common Mistakes
- **Key Takeaways**

- While multiple sets of industry standards exist, there are some **universal rules and principles** which most modellers agree upon
- **Be consistent**, keep models **clear and simple**, distinguish between your **inputs**, **calculations and outputs**, ensure you have an agreed **scope**, and **thoroughly test** models before use
- **Periodicity** is a critical component of virtually every model, and can vary in complexity based on the project context
- Standards will only get you so far; **stay open-minded**, **keep sight of the big picture**, and **keep learning** to stay sharp

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CASE STUDY: REVENUE FORECAST

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CASE STUDY: BASIC REVENUE FORECAST



From: **Sarah Sweet** *(Bakery Owner)*
Subject: **Considering making the move**

Good afternoon!

I'm beginning to seriously consider signing the lease for the property on High Street, but I'd like to better understand the financial implications.

To help me start planning, could you please forecast potential revenue for the bakery if I choose to make the move? I'd love to see projections for the next four years, if possible.

Thank you!

RevenueForecast.xlsx

Reply Forward

Key Objectives

1. Review the information and assumptions Sarah shared in her Excel file
2. Create a simple model to forecast bakery revenue over the next 4 years
3. Use modelling best practices, including clear inputs, calculations and outputs

HINT:

You may need to incorporate Excel functions like **INDEX**, **MATCH** & **SUMIFS** to reach a solution

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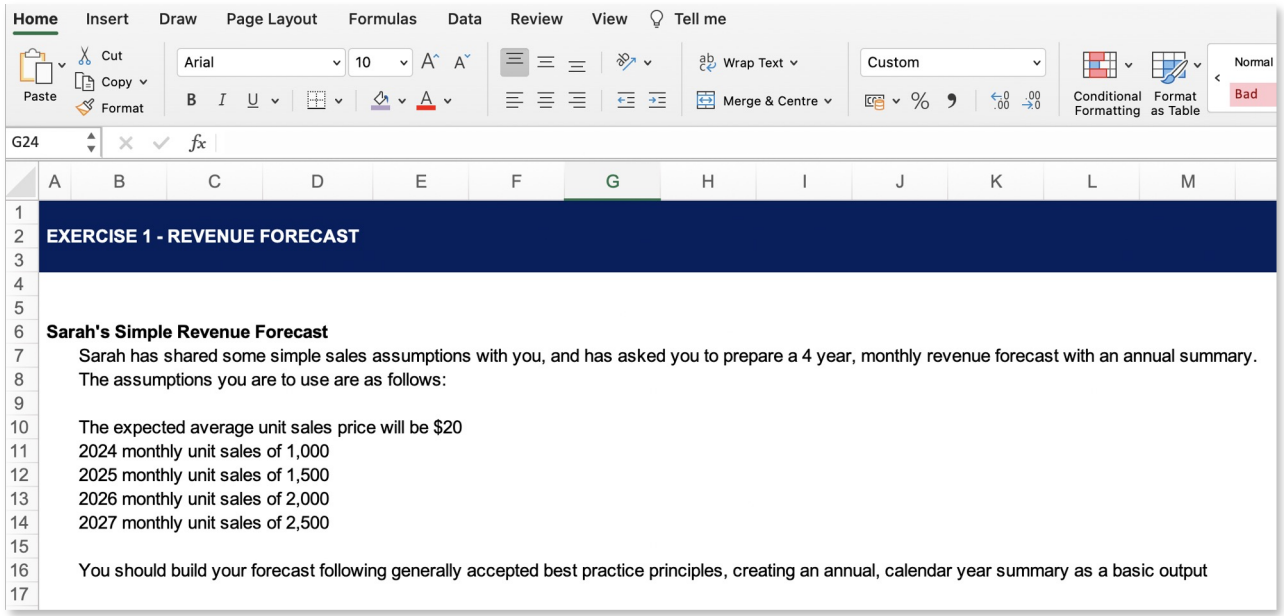
CASE STUDY WORKBOOK

The case study workbook is **completely self-contained**, and includes practice worksheets for you to attempt the case on your own along with solution worksheets to check your work

*These are the **practice worksheets***

*These are the **solution worksheets***





*The **Case Study** tab contains all the information and assumptions you'll need to complete the case*

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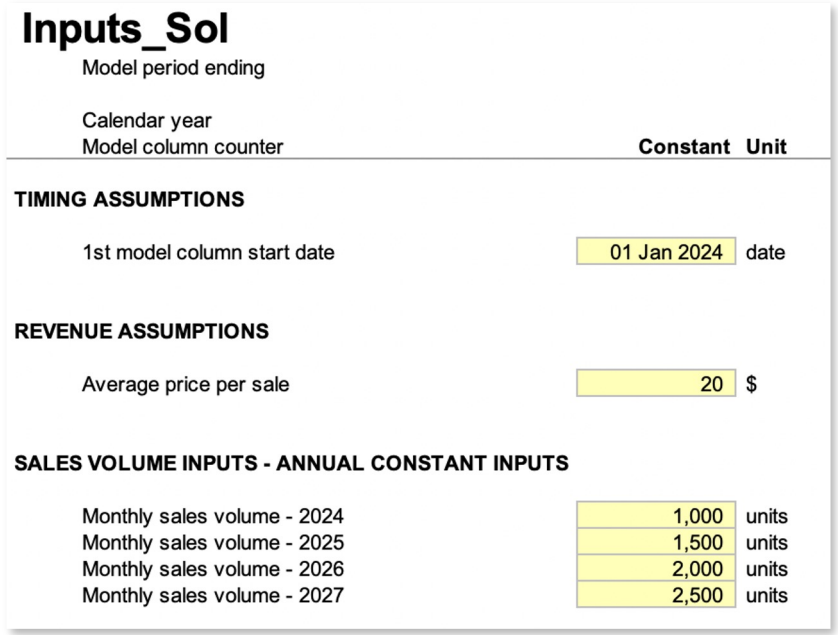
CASE STUDY WORKBOOK

The case study workbook is **completely self-contained**, and includes practice worksheets for you to attempt the case on your own along with solution worksheets to check your work

*These are the **practice worksheets***

*These are the **solution worksheets***





Inputs_Sol

Model period ending		
Calendar year		
Model column counter	Constant	Unit
TIMING ASSUMPTIONS		
1st model column start date	01 Jan 2024	date
REVENUE ASSUMPTIONS		
Average price per sale	20	$
SALES VOLUME INPUTS - ANNUAL CONSTANT INPUTS		
Monthly sales volume - 2024	1,000	units
Monthly sales volume - 2025	1,500	units
Monthly sales volume - 2026	2,000	units
Monthly sales volume - 2027	2,500	units

*The **Inputs** tab is where you'll define the inputs for your calculations.*

Note how the input cells are clearly and uniquely formatted 👍

*(This screenshot is from the **Inputs_Sol** tab and highlights the inputs you will have to recreate on the **Inputs** tab yourself)*

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Calcs_Sol

Model period ending				31 Jan 24 Forecast	29 Feb 24 Forecast	31 Mar 24 Forecast	30 Apr 24 Forecast
Calendar year				2024	2024	2024	2024
Model column counter	Constant	Unit	Total	1	2	3	4
Revenue							
SOLUTION 1							
Average price per sale	20	$					
Monthly sales volume - 2024	1,000	units					
Monthly sales volume - 2025	1,500	units					
Monthly sales volume - 2026	2,000	units					
Monthly sales volume - 2027	2,500	units					
Model year	-	#	- -	1	1	1	1
Revenue		$	1,080,000	20,000	20,000	20,000	20,000

*The **Calcs** tab is where you'll build out your model calculations.*

There are many ways you could tackle this, so we'll walk through our preferred approach along with some common alternatives.

*(This screenshot is from the **Calcs_Sol** tab and highlights the calculations you will have to recreate on the **Calcs** tab yourself)*

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The case study workbook is **completely self-contained**, and includes practice worksheets for you to attempt the case on your own along with solution worksheets to check your work

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Ann_Sum_Sol

Calendar year	Unit	Total	Forecast 2024	Forecast 2025	Forecast 2026	Forecast 2027
Revenue	$	1,680,000	240,000	360,000	480,000	600,000

*The **Ann_Sum** tab is our model output, or annual forecast for the next four years.*

*(This screenshot is from the **Ann_Sum_Sol** tab and highlights the calculations you will have to recreate on the **Ann_Sum** tab yourself)*

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FLOWS & BALANCES

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FLOWS & BALANCES



In this section we'll introduce the concept of flows and balances, and apply these concepts to Sarah's Bakery and her forecasting requirements

TOPICS WE'LL COVER:

- **Definitions**
- **Forecasting Cash**
- **CAPEX**
- **Key Takeaways**

GOALS FOR THIS SECTION:

- *Define flows and balances and how they are commonly represented in financial models*
- *Review an example of a cash balance for Sarah's Bakery*
- *Discuss flows and balances in the context of capital expenditure (CAPEX)*

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DEFINITIONS: FLOWS & BALANCES

Definitions

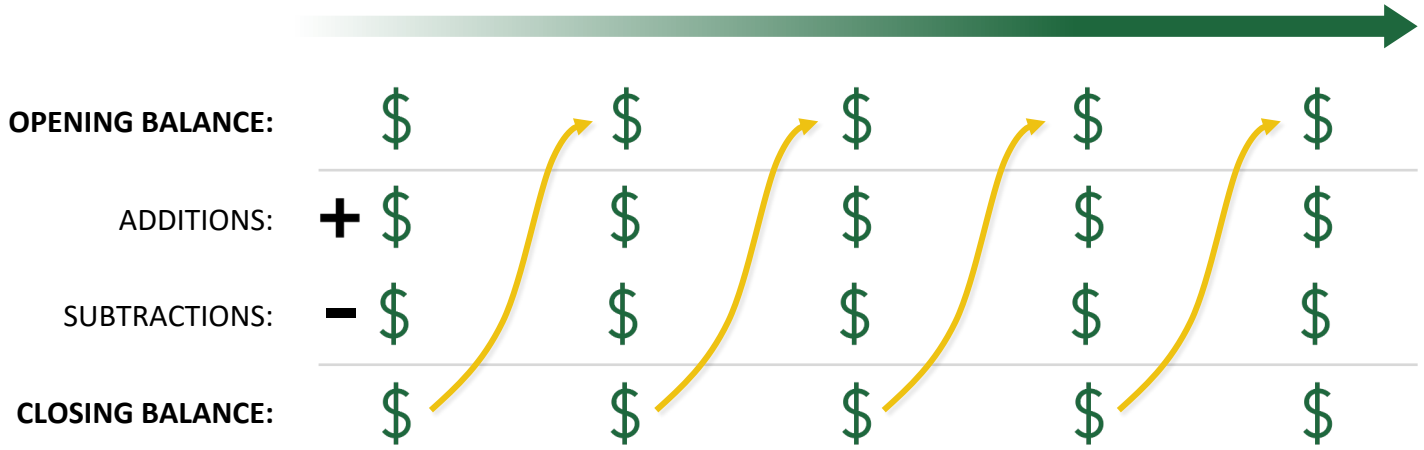
Forecasting Cash

CAPEX

Key Takeaways

Financial models are no more than a collection of Flows & Balances

- **Flows:** positive or negative financial movements (additions/subtractions) within a period
- **Balances:** calculated results (balances) after all flows have been accounted for in a period
- **Financial Models:** layers of flows and balances, linked together with deliberate logic



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CASH FLOWS & BALANCES

Definitions

Forecasting Cash

CAPEX

Key Takeaways

EXAMPLE: Cash flow

	Year 1	Year 2	Year 3	Year 4
Cash Received	100,000	100,000	100,000	100,000
Cash Paid	80,000	90,000	100,000	110,000

EXAMPLE: Cash balance

	Year 1	Year 2	Year 3	Year 4
Cash - Opening Bal	**-**	**20,000**	**30,000**	**30,000**
add Cash Received	100,000	100,000	100,000	100,000
less Cash Paid	80,000	90,000	100,000	110,000
Cash - Closing Bal	**20,000**	**30,000**	**30,000**	**20,000**

BASE: *Beginning, add* ***A****ddition, less* ***S****ubtraction, equals* ***E****nd*
(in modelling terminology, this is known as "The Corkscrew")

Almost all financial models, no matter how complex they may seem, ***are just a series of flows and balances with deliberate logic & relationships holding them all together***

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DEMO: UPDATING SARAH'S REVENUE FORECAST

Let's add some logic to Sarah's model, to **forecast cash receipts (flows),** and the resulting **cash balance**

Calcs_Sol

		Constant	Unit		Total		31 Jan 22 Forecast	28 Feb 22 Forecast	31 Mar 22 Forecast	30 Apr 22 Forecast	31 May 22 Forecast	30 Jun 22 Forecast
	Model period ending											
	End of model year flag						-	-	-	-	-	-
	Calendar year						2022	2022	2022	2022	2022	2022
	Model column counter						1	2	3	4	5	6
Revenue												
	Average price per sale	20	$									
	Monthly revenue - 2022	1,000	units									
	Monthly revenue - 2023	1,500	units									
	Monthly revenue - 2024	2,000	units									
	Model year	-	#		-	-	1	1	1	1	1	1
	Revenue		$		1,080,000		20,000	20,000	20,000	20,000	20,000	20,000
Cash Calculations												
	Cash Receipts											
	Current month	-	%									
	1 month delay	100.00%	%									
	2 Month delay	-	%									
	Revenue	-	$		1,080,000	-	20,000	20,000	20,000	20,000	20,000	20,000
	Model column counter	-	counter		-	-	1	2	3	4	5	6
	Current Month Cash Receipts						-	-	-	-	-	-
	1 Month After Cash Reciepts						-	20,000	20,000	20,000	20,000	20,000
	2 Months After Cash Reciepts						-	-	-	-	-	-
	Total Cash Receipts		$		1,040,000		-	20,000	20,000	20,000	20,000	20,000
	Cash Balance											
	Cash - Opening Balance						-	-	20,000	40,000	60,000	80,000
plus	Total Cash Receipts	-	$		1,040,000	-	-	20,000	20,000	20,000	20,000	20,000
less	[Total Cash Payments]											
	Cash - Closing Balance						-	20,000	40,000	60,000	80,000	100,000

→ ***Flow:*** *Revenue*

→ ***Flow:*** *Cash receipts*

→ ***Balance:*** *Cash*

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CAPITAL EXPENDITURE (CAPEX)

Definitions

Forecasting Cash

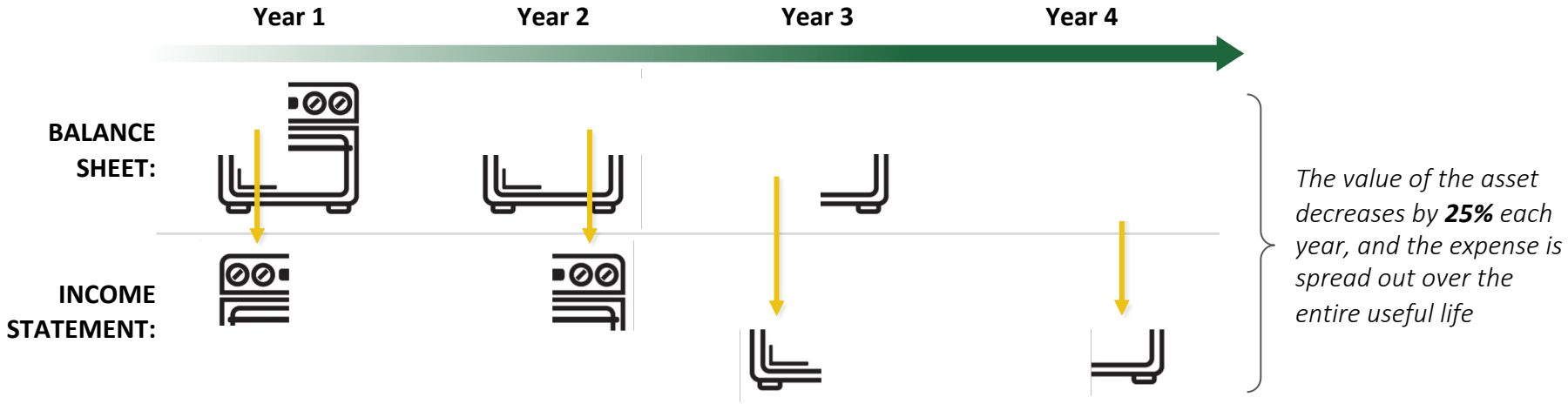
CAPEX

Key Takeaways

Capital expenditure (CAPEX) is the term used to describe money invested by a company on physical assets (buildings, property, equipment, etc.)

- Assets have a **"useful life"**, and the expense is spread out over time as the asset **depreciates**
- The Balance Sheet captures the **Net Book Value (NBV)** of the asset over time, and the Income Statement records how much of the asset's value has been used (or **expensed**) each year

EXAMPLE: Sarah buys a commercial oven, with a useful life of **4 years**



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FORECASTING FLOWS AND BALANCES

Definitions

Forecasting Cash

CAPEX

Key Takeaways

EXAMPLE: Balance Sheet:

		Year 1	Year 2	Year 3	Year 4
	Fixed Assets NBV - BEG	**-**	**30,000**	**20,000**	**10,000**
add	Capex	40,000	-	-	-
less	Depreciation	10,000	10,000	10,000	10,000
	Fixed Assets NBV - END	**30,000**	**20,000**	**10,000**	**-**

EXAMPLE: Income Statement:

	Year 1	Year 2	Year 3	Year 4
Capex	40,000	-	-	-
Depreciation	10,000	10,000	10,000	10,000

***Not all flows and balances relate to cash movements**, and accounting theory plays a critical role here (we'll explore this in more detail in the next section)*

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KEY TAKEAWAYS

Definitions

Forecasting Cash

CAPEX

Key Takeaways

- **Flows** are positive or negative financial movements (additions or subtractions) within a specified time period
- **Balances** are calculated after all relevant flows have been accounted for within a period
- Flows and balances can relate to **cash or non-cash** movements
- **Capital expenditure (CAPEX)** is money invested in physical assets, which have a specific “useful life” and depreciate over time
- Almost all financial models, no matter how complex, can be simplified down to **a series of connected flows and balances**

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CASE STUDY: CAPEX

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CASE STUDY: CAPEX



From: **Sarah Sweet** *(Bakery Owner)*
Subject: **New ovens for the bakery!**

Hi there!

I'm planning to purchase some new commercial ovens for the High Street bakery.

I have estimated how much I'll have to pay for the ovens, and how long I expect them to last (see attached workbook).

Could you please help me work this into our financial model? Thanks!

CapexFlowsandBalances.xlsx

Reply Forward

Key Objectives

1. Review the information and assumptions Sarah shared in her Excel file
2. Create a model to forecast the CAPEX spend, the depreciation charged to the Income Statement, and the Net Book Value of the fixed assets at the end of each year
3. Use modelling best practices, including clear inputs, calculations and outputs

HINT:

You may need to incorporate Excel functions like **AND**, **XLOOKUP** & **SUMIFS** to reach a solution

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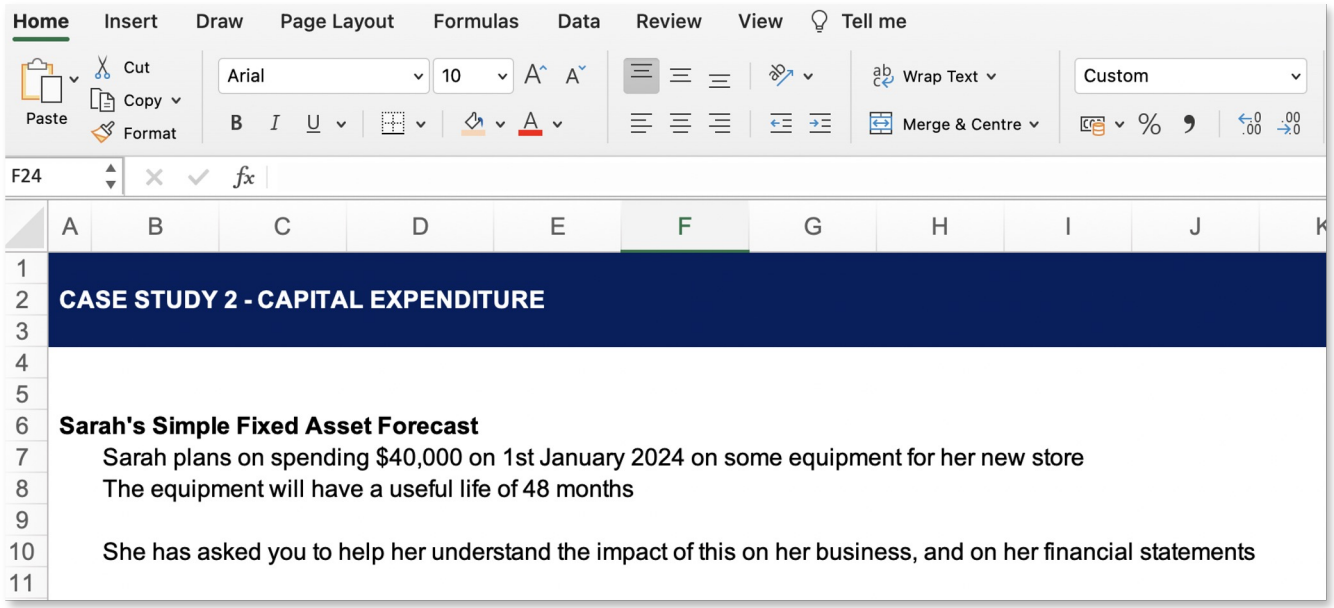
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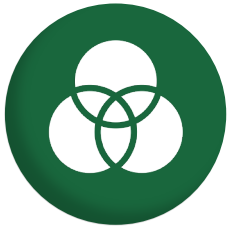
*The **Case Study** tab contains all the information and assumptions you'll need to complete the case*

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THREE STATEMENT MODELLING

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THREE STATEMENT MODELLING



In this section we'll review the accounting equation and key principles, introduce the three core financial statements, and explore how they link in terms of retained earnings and cash

TOPICS WE'LL COVER:

- Accounting Equation
- Key Principles
- The Three Statements
- Balance Sheet
- Income Statement
- Cash Flow Statement
- Retained Earnings
- Key Takeaways

GOALS FOR THIS SECTION:

- *Understand the key accounting equation*
- *Highlight some important accounting principles*
- *Introduce each of the three core financial statements (balance sheet, income statement and cash flow)*
- *Explore how the three statements tie together in terms of retained earnings and cash*

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THE ACCOUNTING EQUATION

- **Accounting Equation**
- Key Principles
- Three Statements
- Balance Sheet
- Income Statement
- Cash Flow Statement
- Retained Earnings
- Key Takeaways

ASSETS = **LIABILITIES** + **EQUITY**

*What a company **OWNS***

*What a company **OWES***

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KEY ACCOUNTING PRINCIPLES

Accounting Equation

Key Principles

Three Statements

Balance Sheet

Income Statement

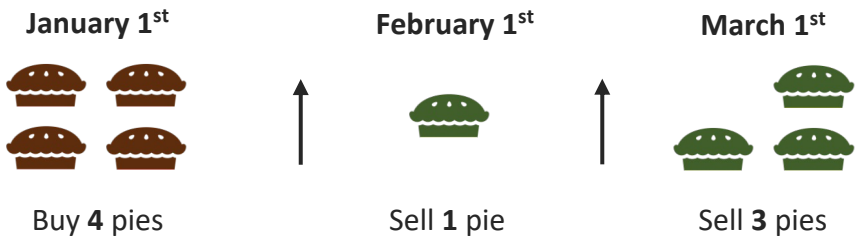
Cash Flow Statement

Retained Earnings

Key Takeaways

PRINCIPLE 1: The **Accruals** Principle

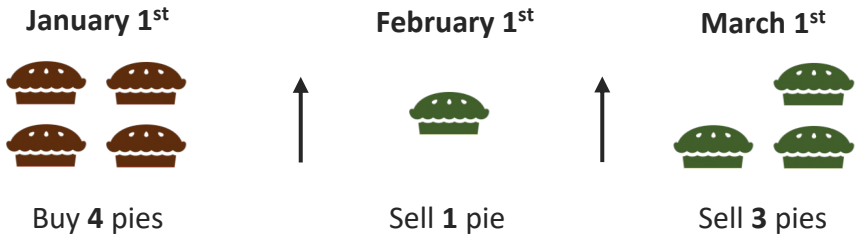
- Revenues are recognised **when services are delivered** (not when related cash is received)



The revenue is "earned" when the products are handed over to the customer *(cash may or may not be exchanged at the time of revenue recognition)*

PRINCIPLE 2: The **Matching** Principle

- Expenses are recognised **alongside related revenues** (not when cash is spent)



An appropriate expense is recognised for the pies ***in the same period the related revenues are recognised***

⚠ ***Cash*** *and* ***profit*** *are not the same!*

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THE THREE FINANCIAL STATEMENTS

Accounting Equation

Key Principles

Three Statements

Balance Sheet

Income Statement

Cash Flow Statement

Retained Earnings

Key Takeaways



BALANCE SHEET

Assets = Liabilities + Equity

A record at a **specific point in time** of everything a companies owns and owes (**Assets, Liabilities** and **Equity**)



INCOME STATEMENT

Revenue - Expenses = ***Profit***

A **summary of a company's profitability** over a period of time (all revenues and expenses related to activity / services delivered)



CASH FLOW

Cash In – Cash Out = ***Net Cash***

A **summary of a company's cash movement** over a period of time (**Operating, Investing** and **Financing** cash flows)

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BALANCE SHEET EXAMPLE

Accounting Equation

Key Principles

Three Statements

Balance Sheet

Income Statement

Cash Flow Statement

Retained Earnings

Key Takeaways

The **balance sheet** captures everything **owned** and **owed** by the company; in other words, it's a record of where funding has come from and how it's been used

BALANCE SHEET					
ASSETS					
Cash and Equivalents	96,000	232,480	438,942	712,842	1,051,108
Accounts Receivable	-	-	-	-	-
Inventory	-	-	-	-	-
Other	-	-	-	-	-
Total Current Assets	**96,000**	**232,480**	**438,942**	**712,842**	**1,051,108**
Net PP&E	30,000	20,000	10,000	-	-
Total Assets	**126,000**	**252,480**	**448,942**	**712,842**	**1,051,108**
LIABILITIES AND EQUITY					
Accounts Payable	-	-	-	-	-
Total Current Liabilities	**-**	**-**	**-**	**-**	**-**
Long Term Debt	50,000	50,000	50,000	50,000	50,000
Total Long Term Liabilities	**50,000**	**50,000**	**50,000**	**50,000**	**50,000**
Total Liabilities	**50,000**	**50,000**	**50,000**	**50,000**	**50,000**
Common Shares	20,000	20,000	20,000	20,000	20,000
Retained Earnings	56,000	182,480	378,942	642,842	981,108
Shareholder's Equity	**76,000**	**202,480**	**398,942**	**662,842**	**1,001,108**
Total Liabilities and Equity	**126,000**	**252,480**	**448,942**	**712,842**	**1,051,108**
Check	-	-	-	-	-

Held / converted to cash < 1 year ← Total Current Assets

Held / useful life > 1 year ← Net PP&E

Expected to pay within 1 year ← Total Current Liabilities

Expected to pay beyond 1 year ← Total Long Term Liabilities

Amount owners have "invested" in the business ← Shareholder's Equity

The crucial Balance Sheet check! ← Check

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SARAH'S BALANCE SHEET

Accounting Equation

Key Principles

Three Statements

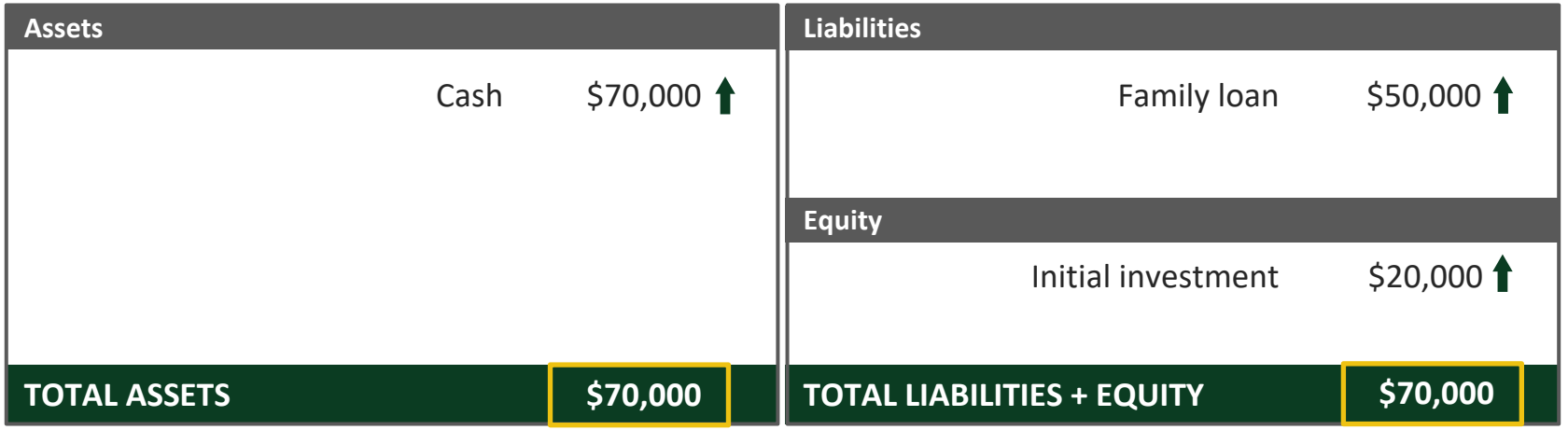
Balance Sheet

Income Statement

Cash Flow Statement

Retained Earnings

Key Takeaways



- **DAY 1:** Sarah starts her business, with **$20,000** of her own money and a **$50,000** loan from family

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SARAH'S BALANCE SHEET

Accounting Equation

Key Principles

Three Statements

Balance Sheet

Income Statement

Cash Flow Statement

Retained Earnings

Key Takeaways

Assets		Liabilities	
Cash	$30,000 ↓	Family loan	$50,000
Fixed Assets	$40,000 ↑	**Equity**	
		Initial investment	$20,000
TOTAL ASSETS	**$70,000**	**TOTAL LIABILITIES + EQUITY**	**$70,000**

- **DAY 1:** Sarah starts her business, with **$20,000** of her own money and a **$50,000** loan from family
- **DAY 2:** Sarah purchases **$40,000** worth of machinery for the shop, with a useful life of **4 years**

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SARAH'S BALANCE SHEET

Accounting Equation

Key Principles

Three Statements

Balance Sheet

Income Statement

Cash Flow Statement

Retained Earnings

Key Takeaways

Assets		Liabilities	
Cash	$30,000	Family loan	$50,000
Fixed Assets	$40,000	Accounts payable	$10,000 ↑
Inventory	$10,000 ↑	**Equity**	
		Initial investment	$20,000
TOTAL ASSETS	**$80,000**	**TOTAL LIABILITIES + EQUITY**	**$80,000**

- **DAY 1:** Sarah starts her business, with **$20,000** of her own money and a **$50,000** loan from family
- **DAY 2:** Sarah purchases **$40,000** worth of machinery for the shop, with a useful life of **4 years**
- **DAY 7:** Sarah purchases **$10,000** of inventory (pie ingredients), which she'll pay for in **7 days**

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SARAH'S BALANCE SHEET

Accounting Equation

Key Principles

Three Statements

Balance Sheet

Income Statement

Cash Flow Statement

Retained Earnings

Key Takeaways

Assets		Liabilities	
Cash	$20,000 ↓	Family loan	$50,000
Fixed Assets	$40,000	Accounts payable	$- ↓
Inventory	$10,000	**Equity**	
		Initial investment	$20,000
TOTAL ASSETS	**$70,000**	**TOTAL LIABILITIES + EQUITY**	**$70,000**

- **DAY 1:** Sarah starts her business, with **$20,000** of her own money and a **$50,000** loan from family
- **DAY 2:** Sarah purchases **$40,000** worth of machinery for the shop, with a useful life of **4 years**
- **DAY 7:** Sarah purchases **$10,000** of inventory (pie ingredients), which she'll pay for in **7 days**
- **DAY 14:** Sarah pays off the inventory balance of **$10,000**

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SARAH'S BALANCE SHEET

Accounting Equation

Key Principles

Three Statements

Balance Sheet

Income Statement

Cash Flow Statement

Retained Earnings

Key Takeaways

Assets		Liabilities	
Cash	$20,000	Family loan	$50,000
Fixed Assets	$30,000 ↓	Accounts payable	$-
Inventory	$10,000	**Equity**	
		Initial investment	$20,000
		Retained Earnings	($10,000) ↓
TOTAL ASSETS	**$60,000**	**TOTAL LIABILITIES + EQUITY**	**$60,000**

- **DAY 1:** Sarah starts her business, with **$20,000** of her own money and a **$50,000** loan from family
- **DAY 2:** Sarah purchases **$40,000** worth of machinery for the shop, with a useful life of **4 years**
- **DAY 7:** Sarah purchases **$10,000** of inventory (pie ingredients), which she'll pay for in **7 days**
- **DAY 14:** Sarah pays off the inventory balance of **$10,000**
- **DAY 365:** The Net Book Value of Sarah's fixed assets reduce by **$10,000** (25%) through straight line depreciation

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INCOME STATEMENT EXAMPLE

Accounting Equation

Key Principles

Three Statements

Balance Sheet

Income Statement

Cash Flow Statement

Retained Earnings

Key Takeaways

The **income statement** is a summary of a company's financial performance, and records **revenues, expenses** and **net income/profit** within a time period

- **NOTE:** This is based on when revenue is earned, NOT when cash is received *(Accruals Principle)*

	INCOME STATEMENT	Year 1	Year 2	Year 3	Year 4	Year 5
Recognised when earned, not when cash is received	Revenue	240,000	360,000	480,000	600,000	720,000
	Total Revenue	**240,000**	**360,000**	**480,000**	**600,000**	**720,000**
	Cost of Sales	(60,000)	(90,900)	(122,412)	(156,075)	(191,036)
Revenue less direct cost of sales	**Gross Profit**	**180,000**	**269,100**	**357,588**	**443,925**	**528,964**
	Operating Expenses	(100,000)	(101,000)	(102,010)	(104,050)	(106,131)
Earnings Before Interest, Tax, Depreciation & Amortisation	**EBITDA**	**80,000**	**168,100**	**255,578**	**339,875**	**422,833**
	Depreciation	(10,000)	(10,000)	(10,000)	(10,000)	-
Earnings Before Interest and Tax	**EBIT**	**70,000**	**158,100**	**245,578**	**329,875**	**422,833**
	Interest Expense					
Earnings Before Tax	**EBT**	**70,000**	**158,100**	**245,578**	**329,875**	**422,833**
	Tax Expense	(14,000)	(31,620)	(49,116)	(65,975)	(84,567)
	Total Tax	**(14,000)**	**(31,620)**	**(49,116)**	**(65,975)**	**(84,567)**
The "bottom line" we need	**Net Income**	**56,000**	**126,480**	**196,462**	**263,900**	**338,266**

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CASH FLOW EXAMPLE

Accounting Equation

Key Principles

Three Statements

Balance Sheet

Income Statement

Cash Flow Statement

Retained Earnings

Key Takeaways

The **cash flow statement** is a summary of cash generated by a company within a specific time period, broken down into **operating, investing** and **finance** activities

*Cash flows generated from the **operations** of the business*

*Cash flows from the **investing** activities of the business*

*Cash flows from all **financing** activities for the business*

The final balance we need

CASH FLOW STATEMENT	Year 1	Year 2	Year 3	Year 4	Year 5
Operating Activities					
Net Income	56,000	126,480	196,462	263,900	338,266
add Depreciation	10,000	10,000	10,000	10,000	-
Changes in Working Capital	-	-	-	-	-
Cash Flows from Operating Activities	**66,000**	**136,480**	**206,462**	**273,900**	**338,266**
Investing Activities					
Capital Expenditure	(40,000)	-	-	-	-
Cash Flows from Investing Activities	**(40,000)**	**-**	**-**	**-**	**-**
Financing Activities					
Debt Issuance / (Repayment)	50,000	-	-	-	-
Equity	20,000	-	-	-	-
Dividends					
Cash Flows from Financing Activities	**70,000**	**-**	**-**	**-**	**-**
Cash at Start of Period	-	96,000	232,480	438,942	712,842
Change in Cash Position	96,000	136,480	206,462	273,900	338,266
Cash at End of Period	**96,000**	**232,480**	**438,942**	**712,842**	**1,051,108**

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RETAINED EARNINGS & CASH

Accounting Equation

Key Principles

Three Statements

Balance Sheet

Income Statement

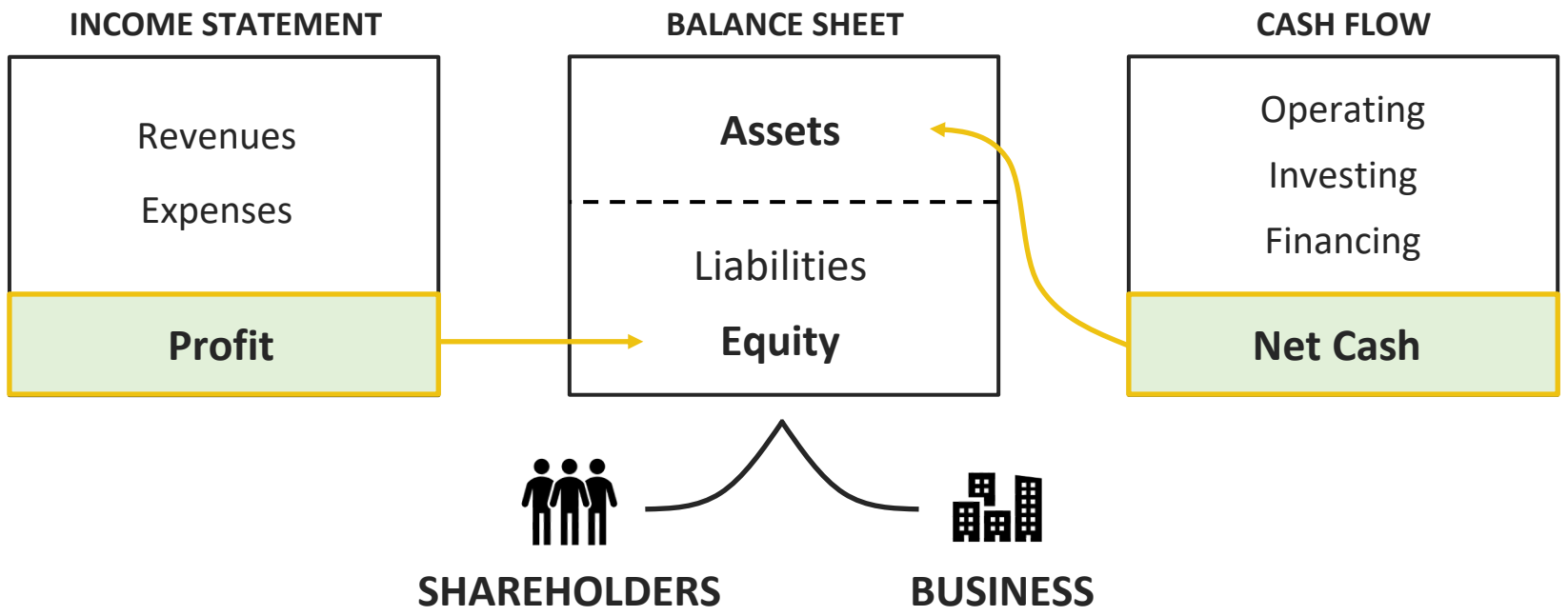
Cash Flow Statement

Retained Earnings

Key Takeaways

Retained earnings and **net cash** tie the three financial statements together

- **Profit** from the income statement flows to the balance sheet as **equity**
- **Net cash** from the cash flow statement flows to the balance sheet as an **asset**
- The company then decides to either **distribute earnings to shareholders** (as dividends), or **reinvest earnings back into the business**



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SARAH'S STATEMENTS (**END OF YEAR 1**)

- Accounting Equation
- Key Principles
- Three Statements
- Balance Sheet
- Income Statement
- Cash Flow Statement
- **Retained Earnings**
- Key Takeaways

CASH FLOW STATEMENT - INDIRECT	Year 1
Net Income	$56,000
Add: Depreciation	$10,000
Less: Change in WC	-
Cash Flows From Operating Activities	**$66,000**
Fixed Assets	($40,000)
Cash Flows From Investing Activities	**($40,000)**
Sarah's Investment	$50,000
Loan	$20,000
Cash Flows From Financing Activities	**$70,000**
Cash Balance - Start	-
Net Cash Movement	$96,000
Cash Balance - End	**$96,000**

INCOME STATEMENT	Year 1
Revenue	$240,000
COS	($60,000)
Gross Profit	**$180,000**
Expenses	($100,000)
EBITDA	**$80,000**
Depreciation	($10,000)
EBIT	**$70,000**
Interest	-
EBT	**$70,000**
Tax (20%)	($14,000)
Net Income	**$56,000**

Assets	End of Year 1	Liabilities	End of Year 1
Cash	$96,000	Loan from family	$50,000
Fixed Assets	$30,000		
		Equity	
		Sarah's Initial Investment	$20,000
		Retained Earnings	$56,000
TOTAL ASSETS	**$126,000**	**TOTAL LIABILITIES + EQUITY**	**$126,000**

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ASSIGNMENT: AMAZON'S 2021 ANNUAL REPORT



amazon

ASSIGNMENT:

Download **Amazon's 2021 Annual Report**, review the financial statements included, and take notes on what they tell you about the business

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ASSIGNMENT: AMAZON'S 2021 ANNUAL REPORT

AMAZON.COM, INC. CONSOLIDATED STATEMENTS OF OPERATIONS
(in millions, except per share data)

Year Ended December 31,	2019	2020	2021
Net product sales	$ 160,408	$ 215,915	$ 241,787
Net service sales	120,114	170,149	228,035
Total net sales	**280,522**	**386,064**	**469,822**
Total operating expenses	265,981	363,165	444,943
Operating income	**14,541**	**22,899**	**24,879**
Total non-operating income (expense)	(565)	1,279	13,272
Income before income taxes	**13,976**	**24,178**	**38,151**
Provision for income taxes	(2,374)	(2,863)	(4,791)
Equity-method investment activity, net of tax	(14)	16	4
Net income	**$ 11,588**	**$ 21,331**	**$ 33,364**

AMAZON.COM, INC. CONSOLIDATED STATEMENTS OF CASH FLOWS
(in millions)

Year Ended December 31,	2019	2020	2021
CASH, CASH EQUIVALENTS, AND RESTRICTED CASH, BEGINNING OF PERIOD	**$ 32,173**	**$ 36,410**	**$ 42,377**
Net cash provided by (used in) operating activities	38,514	66,064	46,327
Net cash provided by (used in) investing activities	(24,281)	(59,611)	(58,154)
Net cash provided by (used in) financing activities	(10,066)	(1,104)	6,291
Net increase (decrease) in cash, cash equivalents, and restricted cash	**4,237**	**5,967**	**(5,900)**
CASH, CASH EQUIVALENTS, AND RESTRICTED CASH, END OF PERIOD	**$ 36,410**	**$ 42,377**	**$ 36,477**

What insights were you able to extract from Amazon's financial statements?

*At the highest level, **Amazon is becoming increasingly more profitable**, generating **$33.4 billion** in Net Income in 2021 (up **56%** from 2020)*

*Amazon **generates a lot of cash** and **invests heavily**; cash used for investing activities **more than doubled in 2020**, and remained at roughly the same level in 2021*

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ASSIGNMENT: AMAZON'S 2021 ANNUAL REPORT

AMAZON.COM, INC. CONSOLIDATED BALANCE SHEETS (in millions, except per share data)	Yesr End Dec 31, 2020	Yesr End Dec 31, 2021
Current assets:		
Cash and cash equivalents	$ 42,122	$ 36,220
Marketable securities	42,274	59,829
Inventories	23,795	32,640
Accounts receivable, net and other	24,542	32,891
Total current assets	**132,733**	**161,580**
Property and equipment, net	113,114	160,281
Operating leases	37,553	56,082
Goodwill	15,017	15,371
Other assets	22,778	27,235
Total assets	**$ 321,195**	**$ 420,549**
LIABILITIES AND STOCKHOLDERS' EQUITY		
Current liabilities:		
Accounts payable	$ 72,539	$ 78,664
Accrued expenses and other	44,138	51,775
Unearned revenue	9,708	11,827
Total current liabilities	**126,385**	**142,266**
Long-term lease liabilities	52,573	67,651
Long-term debt	31,816	48,744
Other long-term liabilitiesCommitments and contingencies (Note 7)	17,017	23,643
Outstanding shares — 503 and 509	5	5
Treasury stock, at cost	(1,837)	(1,837)
Additional paid-in capital	42,865	55,538
Accumulated other comprehensive income (loss)	(180)	(1,376)
Retained earnings	52,551	85,915
Total stockholders' equity	93,404	138,245
Total liabilities and stockholders' equity	**$ 321,195**	**$ 420,549**

What does each balance and YoY movement tell us?

At the highest level, ***Amazon's total assets value has increased by almost 31% from 2020 to 2021***

Despite Amazon's positive operating cash flow position, ***the company has seen an increase in debt and equity funding from 2020 to 2021*** *(i.e. the company's liabilities have increased)*

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KEY TAKEAWAYS

Accounting Equation

Key Principles

Three Statements

Balance Sheet

Income Statement

Cash Flow Statement

Retained Earnings

Key Takeaways

- The **balance sheet** captures everything **owned** and **owed** by the company at a point in time, where **Assets** always equal **Liabilities + Equity**
- The **income** and **cash flow** statements record financial activity over a specific time period
- **Retained earnings/profit & net cash** are the two key links between the income Statement, cash flow statement, balance sheet
- Each financial statement tells you something about the company's past or expected future performance, but **all three together present the full story**

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CASE STUDY: THREE STATEMENT MODEL

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CASE STUDY: THREE STATEMENT MODEL



From: **Sarah Sweet** *(Bakery Owner)*

Subject: **HELP! My model isn't working!**

Good morning!

An old colleague of mine had been helping me develop a new monthly forecast model for the bakery, based on some updated assumptions.

Unfortunately, he couldn't finish the job and I've been left with an incomplete model.

Could you please help me wrap this up?

MonthlyThreeStatementModel.xlsx

Reply Forward

Key Objectives

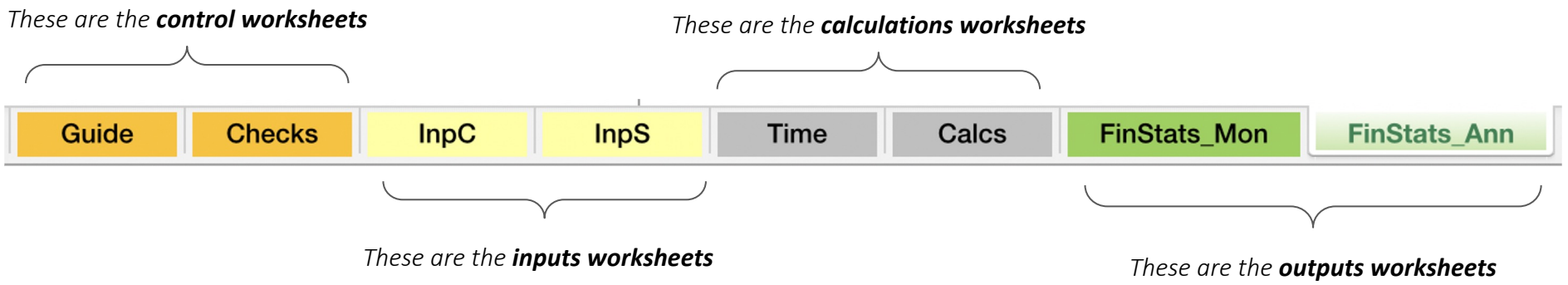
1. Review the financial model provided by Sarah
2. Link up the three statements by completing the two balance corkscrews for Retained Earnings and Cash
3. Highlight any insights that you feel are important for Sarah to know once the model is fully configured

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CASE STUDY WORKBOOK

For this case study, the workbook is set up like a **standalone financial model** rather than a training aid

- **This is just one possible approach to the model structure**; we will discuss this version along with some alternative approaches as we walk through the exercise



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RECAP & NEXT STEPS

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KEY COURSE TAKEAWAYS

- While definitions vary, all financial models should **serve a clear purpose** and **help stakeholders make informed, data-driven decisions**
- While multiple sets of industry standards exist, there are some **universal rules and principles** which most modellers agree upon
- **Be consistent**, keep models **clear and simple**, distinguish between your **inputs**, **calculations and outputs**, ensure you have an agreed **scope**, and **thoroughly test** models before use
- Almost all financial models, no matter how complex, can be simplified down to **a series of connected flows and balances**
- Three statement models consist of balance sheets, income statements and cash flow statements, and **together present the full story about a company's financial performance**
- Standards will only get you so far; **stay open-minded**, **keep sight of the big picture**, and **keep learning** to stay sharp

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NEXT STEPS

This is just the tip of the iceberg! Future topics to explore include:

Scoping, documenting, reviewing and **managing** models once they are built

Intermediate and **advanced modelling techniques:** valuations, M&A, funding, Monte Carlo, etc.

Soft skills: relationship management, presentation skills, communication, etc.

Recommended next steps:

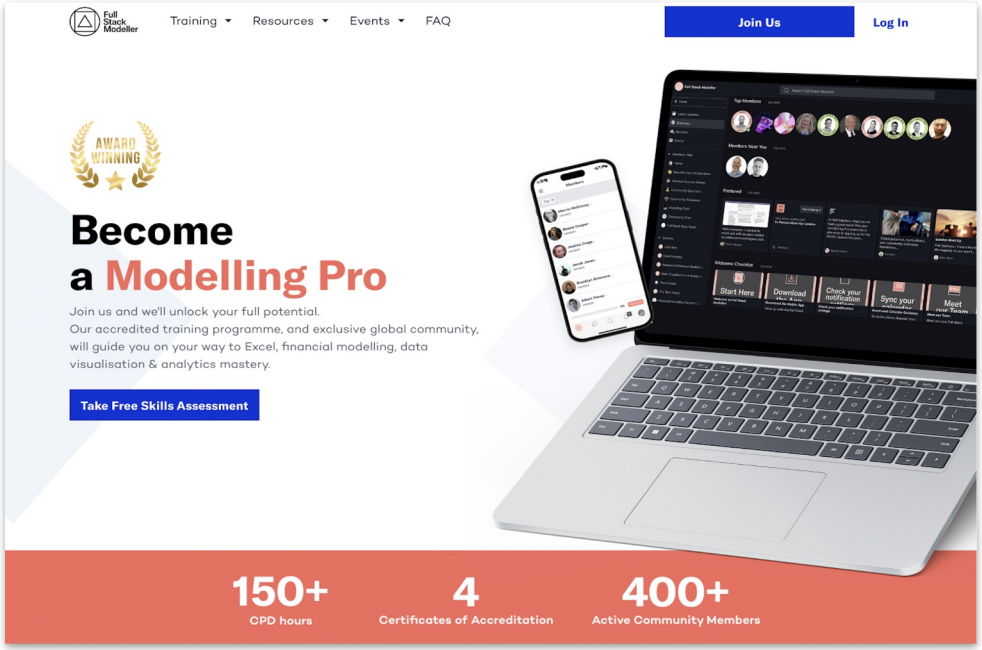
Seek out opportunities to **explore real-world models** and **hone your own modelling skills**

Review and familiarize yourself with **common standards** (ICAEW, FAST, SMART, BPM, etc.)

Stay up to date with **Excel's latest features** (Power Query, dynamic arrays, LAMBDAs etc.)

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